ThoughtSpot — Data Connections, the Semantic Layer, and Dataset Formatting

Research pass for the Spiff design team. All UI labels are quoted verbatim from ThoughtSpot Cloud public docs unless flagged otherwise.

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1. Connections

Sources: docs.thoughtspot.com/cloud/latest/connections · /connect-data · /connections-snowflake-add

31+ connection types. Warehouses: Snowflake, BigQuery, Databricks, Redshift, Athena, Azure Synapse. Relational: PostgreSQL, MySQL, Oracle, SQL Server, Aurora & RDS, Google Cloud SQL. Federation/specialised: ClickHouse, Denodo, Dremio, Presto, Trino, Starburst, SAP HANA, Teradata, SingleStore, Iomete, AlloyDB, Looker, Generic JDBC.

Connect flow (Snowflake, verbatim): Data workspace → Connections → + Create connection → pick tile → Next → Connection name + description → Next → authentication (Use OAuth, OAuth with PKCE, Use External OAuth, OAuth Client Credentials, Key Pair, Service Account, Personal Access Token) → credentials (Account name or Snowflake URL, User, Private Key, Passphrase, Role, Warehouse, Database; an Advanced Config menu adds free-form Key/Value pairs) → Continue → select tables and columns → Create connection (or Save and exit to defer table selection). Primary/foreign keys and existing joins are imported automatically.

Live query vs imported. Connections (historically Embrace) run "live queries on external databases" — no ETL. The one materialised copy: ThoughtSpot "fetches unique values for all attribute data", by default every 24 hours, to power search typeahead. So value suggestions can be stale even when results are live.

Connection health. Docs describe constraints, not a dashboard: "Connections does not support joins across connections"; no column deletion after creation; slow rendering above a few thousand tables. Flag: no per-connection green/red status indicator found in docs — only a Usage "System Liveboard displaying connection statistics."

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2. Models (the semantic layer)

Sources: /semantic-layer · /models-simplify.html · /models.html · /tables-join.html · /join-add · thoughtspot.com/product/spotter-semantics

A Model is "a logical view created on top of a more complex data model, to enable business users to more easily consume data." Models superseded Worksheets; docs call them "the primary data object for search and analytics in ThoughtSpot."

The editor is tabbed — a good IA in itself: Tables (drag in, join) · Columns (pick + configure) · Formulas · Filters (persistent, model-scoped) · Parameters · Instructions (natural-language AI rules) · Settings.

Three source paths: Build your own with cloud data, dbt, TML. Settings carries "Data model join rule" — Apply joins progressively (recommended for most cases) vs Apply all joins — plus an option to "disable row level security for your data model."

Joins. Columns tab → Joins tab → + Add join → the Create Join dialog: Join name, Table 1, Table 2, matched column pairs (+Add column for

composite keys), join type (INNER, FULL OUTER, LEFT OUTER, RIGHT OUTER), cardinality (Many:1, 1:Many, 1:1). Non-equi joins (<, >, <=, >=, !=) are Beta with All/Any matching; range joins are TML-only. Quoted best practice: "creating a many-to-one join from a fact table to a dimension table simplifies your search."

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3. Column-level configuration

Sources: /data-modeling-settings · /tml-properties · /data-modeling-index · /data-modeling-patterns · /data-modeling-geo-data · /data-modeling-aggreg-additive · /data-modeling-attributable-dimension · /spotter-ai-context

Columns are edited in a spreadsheet-like grid with ALL-CAPS property headers (COLUMN NAME, GEO CONFIG, INDEX PRIORITY, CALENDAR TYPE…). Double-click a cell to edit, then Save Changes; bulk edit via the Edit menu. Full list in the inventory below; the notable ones:

- Column Type — ATTRIBUTE ("a characteristic or trait… such as name, address, or id number") or MEASURE ("a numeric value that can be compared… like sales"). Attributes drive the x-axis, measures the y-axis.
- Additive (YES/NO) gates aggregations on numeric attributes. NO → only UNIQUE COUNT OF, TOTAL COUNT OF. YES → also TOTAL OF, AVG OF, STD DEVIATION OF, VARIANCE OF, MIN OF, MAX OF. Note the user-facing names are English phrases, not SQL functions.
- Synonyms — "synonyms that can be used in the search bar to refer to a column." A plain list; the cheapest discovery feature in the product.
- Index Priority — integer 1–10, default 1. Docs recommend 8–10 for important columns, 1–3 for low priority. "Influence[s] the ranking of the column's name" and feeds usage based ranking (UBR).
- Index Type — columns above 100,000 unique values won't index under DEFAULT; only PREFIX_ONLY is recommended at high cardinality.
- Format Pattern — Java notation: #,###, #,##0.## (12345.6789 → 12,345.68), #%, #.00%, MM/dd/yyyy, MM/dd/yyyy HH:mm, MMM.
- Currency Type — the Specify Currency Type dialog gives exactly three options: Infer From Browser, From a column (VARCHAR of ISO codes), Specify ISO Code.
- Geo Config — the Specify Geographic Configuration dialog: Latitude, Longitude, Zip Code, US States, US Counties, Countries, international sub-nation regions, Custom map. Since 9.4.0.cl an Auto select classifier detects geo columns.
- Attribution Dimension (YES/NO) only matters over a chasm trap — "two or more fact tables have no direct relationship to each other except through shared dimensions." Docs admit their own bug: "this setting is at the table level rather than the column level. Note that the current UI implies this is a column-level setting."
- AI Context — a field "next to the Column Description field", max 400 characters (auto-gen targets 150–250), created via AI Context > Generate AI Context. Disambiguates similar columns, encodes rules ("exclude nulls"), explains non-standard formats.

Formulas (/model-formula.html): Formulas tab → Add formula. The editor colour-codes tokens — blue for "formula operators and functions", purple for "the names of columns", black for constants. An Advanced settings gear overrides the output's Data type, measure-or-attribute, and Default Aggregation. A separate "Formula function reference" page holds the function library (not enumerated in this pass).

Custom calendars (/connections-cust-cal-create): Data workspace → Utilities → Custom calendar. Fields: Calendar Name, Connection, Database, Schema; method (Create, Upload File, Existing Table); Table, Start Date/End Date (MM/DD/YYYY), Calendar Type (MONTH_OFFSET, FOUR_FOUR_FIVE, FOUR_FIVE_FOUR, FIVE_FOUR_FOUR), Monthly offset, Start day of week, Quarter name prefix (Q), Year name prefix

(FY). Applied per column via the CALENDAR TYPE cell on a DATE/DATE_TIME column.

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4. Row-level security

Source: /security-rls-implement

RLS is defined on the table, not the model: Data workspace → double-click a table → Row security tab → + Add row security → the Rule Builder. Each rule has a Rule Name and an expression — "an expression that gets evaluated for every row and group combination" — using ts_groups and ts_username. Operators in, =, !=; a green indicator signals validity; aggregate functions unsupported. Rules propagate automatically to derived Models and Answers; a FALSE result shows No data to display. Relevant to Spiff's "runs as its owner" invariant: define once at the table, inherit downstream, with one explicit model-level escape hatch that is off by default.

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5. Data quality & trust

Sources: /liveboard-verify · thoughtspot.com/blog/verified-liveboards · /search-data-refresh-time · /data-source-delete · /system-model · /spotter-model · /data-model-instructions · /spotter-business-terms · /spotter-reference-questions

Verified Liveboards — the clearest certification pattern here. Admins grant a Can verify privilege; holders are verifiers (the blog calls them Data Stewards). Flow: author's ellipses menu → Request verification (optional note; pick which stewards to notify — they get an email with a direct link), or a steward self-marks via Mark as verified; the verifier clicks Approve or Decline. Result: "a blue verified label will be added to the top left of the Liveboard", and the icon "appears on the home page and the Liveboards page." Show Liveboard details reveals verifier name and date. Critically, editing breaks the seal: "a yellow warning banner… indicating it has been edited after verification." Managed at Data → Liveboard verification (verified / pending / rejected). Flag: no equivalent "verified Model" badge found in public docs.

Freshness. "Last data refresh time" is a hover tooltip, not persistent chrome — on a data source name, a column header, or a column in an Answer. It also shows "when it was created and by whom" and, at column level, sample values from that column.

Dependents / impact. Every data source has a Dependents tab listing "the names of the dependent objects (Models and Liveboards), and the columns they use." Deleting a source with dependents raises "Cannot delete because of dependent objects" with each dependent as a clickable link.

Usage & popularity is system data, not inline badges: Product Usage ("what existing Models, tables, and views users search on"), TS: BI Server, TS: AI and BI Stats, TS: Login Activity Tracking, TS: Provisioned Users, TS: Object tracking (Beta). Ranking uses UBR.

Model readiness. A "Spotter optimization" tab surfaces recommendations, not a score: Enable indexing (with View suggestions), Fix date value issues, Fix column type mismatches. Flag: no numeric health/quality score documented.

Governed definitions, three overlapping mechanisms:
- Data model instructions — up to 10,000 characters of global rules per model, from the caret beside the data source name in the search bar or the model's Instructions tab. Doc example: "When calculating revenue, always exclude transactions where Account_Type = 'Internal Test'." Caveat: they do "not automatically override a user's direct query if they conflict."
- Business terms — "a global, reusable definition that maps a word or phrase to: A

specific column or value. A filter. A calculation or formula." At Data workspace → Spotter memory → Memory sources → Business terms → Add business term. Saved at user level unless you have model-editor rights, when it goes global. Now legacy, as are reference questions (saved Q&A pairs via + Add to Coaching under an Answer).

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6. Catalogue & discovery

Source: /data-workspace · /spotter-getting-started · /spotter-capabilities

Data workspace left nav: Data objects (Tables / Models / Views), Connections, Analyst Studio, Usage, Utilities (Import/Export TML, Business data model editor, Schema viewer, Liveboard schedules, Custom calendar, dbt Integration).

The object list is a table: Name · Author · Type · Last modified · Tags. Filters: Source type (all, Models only, tables only, views and SQL views only), Tag, Author, plus name search. Sort by Author / Name / Last modified. Checkbox multi-select enables share, delete, apply/remove tags, export as TML, edit as TML.

Definitions users see: Tables = "Raw tables imported or linked from a connected data source"; Models = "The primary data object for search and analytics"; Views = "Saved SQL-based definitions that produce a virtual table."

"What can I ask?" is answered conversationally, not as a catalogue page — users ask "Give me a quick overview of this dataset. What is this data about?". Transparency: Show work ("How Spotter interprets your question") and Query tokens that "represent the simplified query, and show how the data in the answer was computed" (hover a measure for a natural-language description; hover a filter for an explanation plus the value filtered). Classic search adds Query details, Query visualizer, Query SQL. Feedback is Is this useful? (check / X). On cross-model search Spotter shows "a percentage score of its confidence in each choice and the reason it chose them."

A dataset's full property inventory

Connection level — Connection name · Description · Connection type · Authentication type (OAuth / OAuth PKCE / External OAuth / OAuth Client Credentials / Key Pair / Service Account / Personal Access Token) · Account or URL · User · Role · Warehouse · Database · Schema · Private Key / Passphrase · Advanced Config key–value pairs · Selected tables & columns · Imported primary/foreign keys · Attribute-value refresh cadence (default 24h)

Table level — name · description · db · schema · db_table · connection (GUID) · guid · joins (name, Table 1, Table 2, column pairs, type, cardinality) · rls_rules (rule name + expression using ts_groups / ts_username) · Attribution Dimension (YES/NO, table-scoped despite column-level UI) · Tags · Author · Last modified

Model level — name · description · Tags · Author · Created by/on · Last modified · tables · joins · join_path · join_progressive (Apply joins progressively vs Apply all joins) · is_bypass_rls (default false) · parameters · persistent Filters · column_groups (organises the search panel) · Data model instructions (≤10,000 chars) · AI Context · lesson_plans · attached reference questions / business terms · Spotter optimization findings

Column level
- name (Column Name) · description · AI Context (≤400 chars)
- column_id · db_column_name · db_column_properties
- data_type — read only: INT32, INT64, BOOL, VARCHAR, DOUBLE, FLOAT, DATE, DATETIME, TIME
- column_type: ATTRIBUTE | MEASURE

- aggregation: NONE (attribute default), SUM (measure default), AVERAGE, MIN, MAX, STD_DEVIATION, VARIANCE, COUNT, COUNT_DISTINCT
- is_additive: YES/NO
- synonyms: list of alternate search words
- is_hidden: true/false
- index_type: DEFAULT, DONT_INDEX, PREFIX_ONLY, PREFIX_AND_SUBSTRING, PREFIX_AND_WORD_SUBSTRING
- index_priority: 1–10 (default 1)
- Suggestion settings (data-value suggestions on/off)
- format_pattern (numbers and dates)
- currency_type: Infer From Browser | From a column | Specify ISO Code
- geo_config + geometryType (POINT, LINE_STRING, POLYGON, MULTI_POINT, MULTI_POLYGON) + custom_file_guid
- calendar: default (Gregorian) / fiscal / named custom calendar
- spotiq_preference: EXCLUDE
- is_attribution_dimension · is_one_to_one
- Formula columns additionally carry properties → column_type + aggregation
- Derived-but-surfaced: sample values (in tooltip) · last refreshed · created by

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Design lessons for Spiff

1. Make provenance persistent, not a hover. ThoughtSpot's freshness tooltip carries the right payload — last refreshed, created by whom, sample values — but is invisible until you hover. Put those three in permanent chrome on the dataset card.
2. Sample values are the fastest trust signal a non-technical person has. GST-2024-0871 beats any type annotation. Show them wherever a column name appears.
3. Steal the whole verification lifecycle, not just the badge. Request → notify a named steward → Approve/Decline → badge with verifier name and date → yellow "edited since verification" banner. That last step is what makes the badge honest.
4. Certify the thing people share, not the plumbing. ThoughtSpot verifies Liveboards, not Models. Spiff's certifiable unit should be the answer or automation someone forwards.
5. Synonyms are the highest-leverage, lowest-cost field. One box where an owner writes "turnover, revenue, sales" beats any join config for a non-technical searcher.
6. Two axes only: attribute vs measure — "thing you group by" and "number you do maths on." Use the English aggregation names (TOTAL OF, AVG OF, UNIQUE COUNT OF), not SQL.
7. Give every dataset a Dependents view. "3 automations and 2 shared answers use this," with links, turns a scary edit into an informed one.
8. Search ranking is a designable property. Index Priority 1–10 lets an owner say "when someone types 'client', mean this column." Expose it as "Promote in search."
9. Write governed rules in English and store them. Data model instructions (10k chars) and AI Context (400 chars/column) are just prose — and the most modern part of the product. A single agreed metric definition needs no DSL.
10. Two-tier definitions: personal, then promoted. A business term saves at user level unless you have editor rights, when it goes global. That gradient fits Spiff's ownership model.
11. Show the work in tokens. Show work plus hoverable Query tokens describing a measure or filter in natural language is the best "how did you get this number" pattern here.
12. Confidence needs a reason, not just a number. Spotter shows a percentage and why it picked that model.
13. Recommendations beat scores. ThoughtSpot ships "fix these three things" instead of a health grade. Actionable > gradeable.
14. Never render a property at a scope it doesn't belong to. ThoughtSpot documents its own bug: Attribution Dimension is table-level but shown in a column grid.
15. Keep the catalogue list boring and filterable. Name · Author · Type · Last modified · Tags, filtered by source type, tag and author. Add "Verified" and "Last refreshed" as sortable columns and you are ahead.

Screens/components worth stealing

- Column grid as spreadsheet — one row per column, ALL-CAPS property headers, double-click a cell to edit, Save Changes at top, multi-select for bulk edit. Scales to 200 columns with no modals.
- Tabbed model editor — Tables / Columns / Formulas / Filters / Parameters / Instructions / Settings: a clean IA for "everything about this dataset."
- Create Join dialog plus a schema canvas where joins are drawn by dragging a +.
- Verified badge + edited-since-verification banner + Show Liveboard details panel.
- Dependents tab listing dependent objects and the columns they use.
- Freshness tooltip on a source name or column header: last refresh, creator, sample values.
- Specify Currency Type / Specify Geographic Configuration — tiny single-purpose modals launched from one grid cell.
- Formula editor with token colour-coding plus an Advanced settings gear for output type and aggregation.
- Spotter optimization tab — a punch-list of fixable data issues with View suggestions.
- Show work / Query tokens transparency panel with Is this useful? check/X feedback.
- RLS Rule Builder — name + expression, live green validity indicator, ts_groups / ts_username variables.

Flagged as unverified

- No per-connection health/status indicator found in public docs (only a Usage system Liveboard and documented limitations).
- No "verified/certified Model" badge found — verification is documented for Liveboards only.
- No numeric model health or quality score; Spotter readiness is a recommendation list.
- Data workspace tab labels vary between doc versions (10.x vs 26.x) — treat as approximately current, not pixel-exact.
- The formula function reference was not enumerated in this pass.
- lesson_plans and column_groups appear in TML properties but I found no UI documentation for them; treat as code-only until confirmed.